

## Parts H & I: Test Summary Report and Foundations Reflection

### Cinema Distribution System

Document type: Test summary (Part H) and Foundations reflection (Part I)

Group / repository: Enterprise-F-Project / cinema-

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This report summarises the testing campaign for the Cinema Distribution System, compares outcomes to the exit criteria in the test plan, states residual risk, and gives a release recommendation. Part I reflects on one real defect using error / fault / failure and verification versus validation.

## PART H - Test summary report

### H.1 What was tested

- Unit tests of AuthService, MovieService, LicenseService, and RentalService using Mockito test doubles.
- Controller Bean Validation (BVA/EP) for registration, movie duration, and license movied.
- Security integration with MockMvc + Testcontainers PostgreSQL (JWT, roles, blacklist, disabled users).
- System tests with Selenium Page Object Model (client navigation, distributor movie create, rental lifecycle).
- Continuous integration: GitHub Actions workflow and Jenkinsfile running the Maven suite.
- Coverage measurement with JaCoCo on core business services.
- Manual API checks with Postman and demo role accounts.

### H.2 What was not tested (or only lightly)

- Formal load, stress, or soak performance testing.
- Mobile clients and cross-browser matrix beyond Chrome for Selenium.
- Full accessibility (WCAG) certification.
- Changing production SecurityConfig to return 401 (deferred finding DEF-002).
- Blocking ADMIN self-registration in production (deferred finding DEF-001).
- Selenium on CI agents (excluded where frontend/Chrome stack is unavailable; verified locally).

### H.3 Results against exit criteria

| Exit criterion (from Part A)                 | Result | Evidence                                     |
|----------------------------------------------|--------|----------------------------------------------|
| CI suite green (unit+validation+security)    | Met    | GitHub Actions / local mvn; 85 tests CI path |
| Selenium green on full local stack           | Met    | CinemaSystemTest 3/3 BUILD SUCCESS           |
| Core branch coverage >= 90%                  | Met    | JaCoCo 97.3% (71/73) on four services        |
| Defects logged; critical path fixed/deferred | Met    | Parts F/G log; DEF-004 closed                |
| Clear release/demo recommendation            | Met    | Section H.5 below                            |

### H.4 Outstanding defects and residual risk

| ID      | Status   | Residual risk if released as-is                                                    |
|---------|----------|------------------------------------------------------------------------------------|
| DEF-001 | Deferred | Attacker/user can self-register as ADMIN - high security risk for real production. |
| DEF-002 | Deferred | 403 instead of 401 may confuse clients; lower severity than privilege issues.      |
| DEF-003 | Deferred | Disabled users get generic auth failure; messaging less clear.                     |
| DEF-004 | Closed   | No residual rental date-format failure after UI fix and retest.                    |

Residual risk summary: For an academic demonstration with known demo accounts and controlled access, deferred security findings are acceptable if clearly disclosed. For a public internet deployment, DEF-001 must be fixed before release.

### H.5 Quality stopping decision / recommendation

Recommendation: READY for academic release and course demonstration.

Rationale: The test pyramid is in place and green; core coverage exceeds 90%; the blocking rental workflow defect was fixed and re-verified; CI automation exists; remaining defects are documented with severity and priority. Recommendation: NOT READY for unconstrained production internet release until DEF-001 (and preferably DEF-002) are remediated.

### H.6 Key quantitative outcomes

| Measure                                   | Value                                         |
|-------------------------------------------|-----------------------------------------------|
| Full local automated tests (approx.)      | ~88 (unit + validation + security + Selenium) |
| CI automated tests (no Selenium)          | 85                                            |
| Core service branch coverage              | 97.3%                                         |
| Defects logged / closed / deferred        | 4 / 1 / 3                                     |
| Defect removal efficiency (known escapes) | 100%                                          |

## **PART I - Foundations reflection**

Course Unit 1 distinguishes error, fault, and failure, and verification versus validation. The following reflection applies those ideas to one real defect from this project.

### **I.1 Selected defect: DEF-004 (rental date format)**

When a CLIENT created a rental through the web UI, the request failed with HTTP 400 "Invalid value for request field". The Selenium rental-lifecycle scenario could not complete until this was understood and fixed.

### **I.2 Error, fault, and failure**

- Error (human mistake): The team designed the API to accept LocalDateTime while the UI collected only calendar dates (YYYY-MM-DD) and posted them unchanged.
- Fault (defect in the artifact): A contract mismatch between CreateRentalRequest (LocalDateTime fields) and the frontend rentals form payload (date-only strings).
- Failure (observed incorrect behaviour): Jackson rejected the body; the API returned 400; the rental sheet stayed open; users could not complete the rental workflow.

### **I.3 Verification or validation?**

Verification asks whether the product was built correctly against a specification of a component. Our RentalService unit tests verify business rules when given a correctly typed CreateRentalRequest; they would not detect a UI that sends the wrong JSON shape.

Validation asks whether the right product was built for real use. Selenium drove the actual UI against the running API (and we confirmed with Postman/API calls that date-only fails while datetime succeeds). Therefore DEF-004 was caught primarily by validation-style system testing, not by unit-level verification alone.

### **I.4 What we learned**

A green unit suite does not guarantee an end-to-end journey works. Integration and system tests close that gap. After fixing the UI to append T00:00:00, we re-ran Selenium and confirmed the failure mode was gone - moving the defect through Fixed -> Verified -> Closed in the defect lifecycle.

*End of Parts H and I - Test Summary Report and Foundations Reflection.*